give her lots of grandchildren. Longevity is not prima facie evidence of virility. Indeed a long-lived male may have survived precisely *because* he does not take risks in order to reproduce. A female who selects an old male is not necessarily going to have more descendants than a rival female who chooses a young one who shows some other evidence of good genes.

What other evidence? There are many possibilities. Perhaps strong muscles as evidence of ability to catch food, perhaps long legs as evidence of ability to run away from predators. A female might benefit her genes by allying them with such traits, since they might be useful qualities in both her sons and her daughters. To begin with, then, we have to imagine females choosing males on the basis of perfectly genuine labels or indicators which tend to be evidence of good underlying genes. But now here is a very interesting point realized by Darwin, and clearly enunciated by Fisher. In a society where males compete with each other to be chosen as he-men by females, one of the best things a mother can do for her genes is to make a son who will turn out in his turn to be an attractive he-man. If she can ensure that her son is one of the fortunate few males who wins most of the copulations in the society when he grows up, she will have an enormous number of grandchildren. The result of this is that one of the most desirable qualities a male can have in the eyes of a female is, quite simply, sexual attractiveness itself. A female who mates with a super-attractive he-man is more likely to have sons who are attractive to females of the next generation, and who will make lots of grandchildren for her. Originally, then, females may be thought of as selecting males on the basis of obviously useful qualities like big muscles, but once such qualities became widely accepted as attractive among the females of the species, natural selection would continue to favour them simply because they were attractive.

Extravagances such as the tails of male birds of paradise may therefore have evolved by a kind of unstable, runaway process.* In the early days, a slightly longer tail than usual may have been selected by females as a desirable quality in males, perhaps because it betokened a fit and healthy constitution. A short tail on a male might have been an indicator of some vitamin deficiency—evidence of poor food-getting ability. Or perhaps short-tailed males were not very good at running away from predators, and so had had their tails bitten off. Notice that we